



ET AI HACKATHON 2026 · PROBLEM STATEMENT 8

AI for Industrial Knowledge Intelligence

SMRITI

The plant's **institutional memory** — one brain that holds every drawing, work order, procedure, inspection, near-miss and regulation, answers in plain language **with citations you can trust**, and warns you **before** a failure repeats.

स्मृति · Sanskrit for "memory"

THE PROBLEM

Every plant already knows how to prevent its next failure.

That knowledge is trapped in **twelve disconnected systems** and **three retiring engineers' heads**. So the same failure repeats — and nobody connects today's job to the near-miss from 2019.

SILOED

**P&IDs · CMMS · SOPs · incidents
· manuals · email**

None of them talk to each other.

UNDIGITISED

**Drawings & scanned OEM
manuals**

The richest knowledge is locked in
images and PDFs.

TRIBAL

Experience walks out the door

When a senior engineer retires, the
memory retires with them.

WHY NOW — THE MARKET IS PROVEN

A \$3.1B market — just validated.

Schneider Electric / AVEVA acquired **Cognite** for **\$3.1B** (June 2026). The incumbents are heavyweight, Western, enterprise-priced, 9–12-month rollouts.

- **India-first** — real OISD / Factories Act / PESO compliance built in
- **Reads drawings visually** — no OCR, the whole document estate
- **Turns near-misses into warnings** — before the next incident

\$3.1B

Cognite acquisition, Jun 2026

India

The underserved, regulation-heavy market SMRITI targets first

Not another RAG chatbot. A tri-modal operations brain.

One question fans out across **text**, a **knowledge graph**, and the **drawings themselves** — fused, reranked, and answered **on the P&ID with citations**.

01 · FUSION

Text + graph + visual, in one answer

"Why does P-101 trip?" needs drawing connectivity + work-order history + a visually-retrieved OEM page. A text-only RAG can't produce it.

02 · VISION

OCR-free visual retrieval

Every page indexed as an image (colSmol-256M, patch-level MaxSim). Scanned, stamped, even a 1958 legacy drawing — retrieved on visual signal alone.

03 · REASONING

The graph connects what no human does

RCA walks the fleet to surface the monsoon seal pattern across P-101 / P-103 / P-107 — and matches tomorrow's permit to a 2019 near-miss.

04 · TRUST

Every sentence cites; "insufficient evidence" is allowed

Citations click through to the exact page. Faithfulness is measured, not claimed.

HOW IT WORKS — ARCHITECTURE

One query → tri-modal fabric → cited answer on the drawing

SOURCES

P&IDs

Work orders

SOPs (rev chains)

Inspections

Incidents / near-miss

OEM manuals (scanned)

Permits

Email

OISD / Factories Act / PESO

▼ Ingestion — per-type parsers · LLM entity extraction · **continuous intake** (forms · bulk CMMS · auto-classify)

KNOWLEDGE STORE

Industrial Knowledge Graph — typed ontology + provenance

Qdrant · text (dense + BM25)

Qdrant · visual (MaxSim)

DrawingRegion index (pixels ⇔ nodes)

▼ Retrieval — intent router → text || graph || visual → RRF merge → cross-encoder rerank → provenance

AGENTS

1 · Ingestion / KG

2 · Expert Copilot

3 · Diagnostics / RCA

4 · Compliance

5 · Lessons & Proactive Warnings

▼ Presentation — streaming chat · Drawing Viewer w/ live overlays · reasoning trace · dashboards

BUILT TO RUN FOR YEARS, NOT JUST TO DEMO

New knowledge flows in live — and the system reacts instantly.

Log an incident, work order, inspection or permit through a typed form (or bulk CMMS export, or drop a document and it self-classifies). Each writes the **same graph structure** as the core data — so it's live with **no restart, no re-ingest**.

INSTANT MEMORY

Log an incident on P-101

In its Diagnostics timeline seconds later, cited in the next RCA.

LIVE FORESIGHT

Log tomorrow's permit

Matched against the plant's own precursor history → a warning fires on the spot.

AUDITABLE

Stamped manual intake · conf 1.0

Every hand-entered fact carries provenance.

18 real clauses, each with its source — mapped to equipment.

OISD-STD-116 / 129 / 132 / 105 · Factories Act 1948 §§31/36/37/38/40 · SMPV(U) 2016 · Petroleum Rules 2002 — linked to assets by GOVERNED_BY and evaluated against **actual records**.

It catches what humans miss

SMRITI flags that the plant's own inspection plan assumes a **5-year interval** where **OISD-129 cl. 11.1 requires annual** — and that **PSV-1101's test is overdue**. Real gaps, cited to the clause.

PROOF — MEASURED, NOT CLAIMED

SMRITI vs a vanilla single-vector RAG baseline

33-question expert golden set · LLM-judged · same answer model & corpus for both (so deltas are honest)

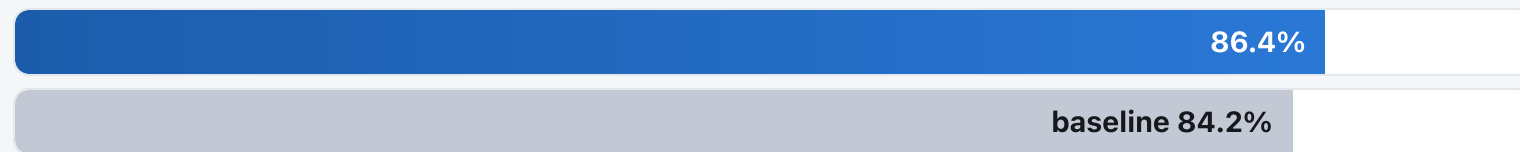
Faithfulness



Citation correctness



Expected-point coverage



Wins concentrate where the fabric earns them: **multi-modal fusion +15 pts, proactive warnings +17 pts, honest partial-evidence +50 pts.**

Two caveats we state, not hide: a text-only judge under-credits the visual answers, and judge shares the answer model family.

■ SMRITI ■ Vanilla RAG

LIVE PRODUCT

Let's open the brain.

A working prototype — ~300-node fabric, 5 agents, running right now.

Ask

Diagnostics

Warnings

Add data → live warning

THE CLOSE

Institutional memory that never retires.

We didn't build a search box. We built the plant's memory — one that reads its drawings, reasons across its history, proves its answers, and taps you on the shoulder before the next failure.

Tri-modal

text + graph + OCR-free visual, fused & cited

India-first

real OISD / Factories Act / PESO, sourced

Proactive

near-misses → warnings before the work

github.com/infinity2147/Bhadra-ET

Prototype live · eval reproducible